MODULE 34 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

Database and Data Migration

Move databases with minimal downtime, convert schemas between engines, and move files and bulk data online, or physically when the network cannot.

DOMAIN 4 of 5 EXAM WEIGHT 15% MODULE 34 of 52

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Copy everything once, then **keep copying the changes** until the moment you switch. Downtime is only the final catch-up.

Homogeneous or heterogeneous?

SAME ENGINE

MySQL to RDS for MySQL

- No schema conversion
- Native tools or DMS homogeneous migrations
- Full load plus replication
- Lowest risk path

DIFFERENT ENGINE

Oracle to Aurora PostgreSQL

- Schemas and code must be converted
- DMS Schema Conversion, then DMS
- Full load plus change data capture
- Test application SQL thoroughly

THE TIEBREAKER

Changing engine adds a conversion step before any data moves.

AWS DMS building blocks

PIECE	ROLE
Schema Conversion	Converts schemas and code; rules first, generative AI for the rest
Full load	Copies existing data
Change data capture	Streams ongoing changes until cutover
Replication instance or DMS Serverless	Runs the task; serverless scales itself
Data validation	Compares source and target rows

ON THE EXAM

Heterogeneous with minimal downtime: Schema Conversion, then full load plus CDC, then validate.

Database migration practices

- **Source prerequisites**: archive logs or binary logs enabled for CDC.
- **Large tables**: parallel full load by partition or range.
- **LOBs** slow tasks; choose limited LOB mode where sizes allow.
- **Secondary indexes and triggers** after full load, before cutover.
- **Cutover**: stop writes, wait for CDC lag zero, validate, switch.

> ON THE EXAM
>
> CDC needs the source's transaction logs available and retained long enough.

Moving files and objects

NEED	SERVICE
NFS, SMB, HDFS, object stores to S3, EFS, FSx	AWS DataSync
Other clouds' storage to S3	DataSync Enhanced mode, agentless
Partners keep sending by SFTP or FTPS	AWS Transfer Family
Ongoing hybrid file access	Storage Gateway (not a migration tool)
Petabytes, network too slow	AWS Data Transfer Terminal

ON THE EXAM

One-off or scheduled bulk copy with verification and incremental passes: DataSync.

When the network is too slow

OPTION	STATUS
AWS Snowball Edge and Snow Family	Closed to new customers
AWS Data Transfer Terminal	Bring your devices to an AWS facility
More bandwidth: Direct Connect upgrade	Weeks to provision
DataSync over existing links	Fine if the time budget allows

ON THE EXAM

A Snowball answer for a new customer is stale. Data Transfer Terminal is AWS's named replacement.

Do the arithmetic

- **1 Gbps** fully used moves roughly 10 TB a day.
- **Real links** rarely sustain more than about 80% of line rate.
- **500 TB at 1 Gbps**: about two months, if nothing else uses the link.
- **2 PB at 1 Gbps**: over half a year. Move it physically.
- **Then sync the deltas** online after the bulk copy lands.

> ON THE EXAM
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> Bulk physically, then catch up the changes online with DataSync.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Oracle to PostgreSQL, minimal downtime	Schema Conversion, DMS full load plus CDC
Same engine, minimal downtime	Homogeneous migration or native replication
Petabytes, slow link, new customer	Data Transfer Terminal
NAS shares to EFS or FSx, then cutover	DataSync incremental tasks
Partners must keep using SFTP	Transfer Family
Variable load, no sizing	DMS Serverless

ON THE EXAM

Engine change means conversion first. Network too slow means physical. Snowball means stale.

> Meridian Freight is moving its 3 TB Oracle shipment database to Aurora PostgreSQL to end Oracle licensing. It has hundreds of PL/SQL procedures.
>
> The application can be offline for no more than 15 minutes at cutover.

What should the solutions architect recommend?

A. AWS DMS full load only, with the application offline during the copy

B. DMS Schema Conversion for schema and code, then a DMS task with full load and change data capture until cutover

C. Rehost the Oracle server to EC2 with AWS Transform MGN

D. Export to CSV, load into Aurora, and rewrite procedures manually

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Full load only

Offline for the whole copy, far beyond 15 minutes.

B Schema Conversion plus full load and CDC

Converts code, then keeps the target current until a short cutover.

C Rehost Oracle

Keeps the Oracle licence the company wants to end.

D CSV and manual rewrite

Long downtime and slow, error-prone conversion.

THE SIGNAL

Engine change, code to convert, minutes of downtime: Schema Conversion, then full load plus CDC.

> Meridian Freight's 5 TB on-premises MySQL 8.0 database must move to Amazon RDS for MySQL 8.0 with less than 10 minutes of downtime.
>
> The team prefers the engine's native replication mechanisms over row-by-row copying.

What should the solutions architect use?

A DMS Schema Conversion to PostgreSQL

B mysqldump with the database offline

C DMS homogeneous data migration, using native MySQL tools for full load and ongoing replication

D AWS DataSync of the MySQL data directory

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Convert to PostgreSQL

Changes engine for no stated reason.

B Offline dump

Hours of downtime for 5 TB.

C Homogeneous migration

Native tools, full load then replication, short cutover.

D Copy data files

Not consistent, and RDS does not accept raw data directories.

THE SIGNAL

Same engine, native mechanisms, short cutover: homogeneous data migration.

> Meridian Freight's media archive of 2 PB of scanned documents sits in a Los Angeles data centre with a 1 Gbps internet link. The archive must be in Amazon S3 within six weeks.
>
> The company has never used AWS Snow Family devices.

What should the solutions architect recommend?

A. Order AWS Snowball Edge devices

B. Copy the archive to drives and upload it at an AWS Data Transfer Terminal, then sync recent changes with DataSync

C. AWS DataSync over the existing 1 Gbps link

D. S3 Transfer Acceleration

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Snowball Edge
Not available to new customers.

B Data Transfer Terminal plus DataSync
Physical bulk upload, then online catch-up.

C DataSync over 1 Gbps
Over half a year for 2 PB.

D Transfer Acceleration
Still limited by the 1 Gbps link.

THE SIGNAL

Too much data for the link, new customer: Data Transfer Terminal, not Snowball.

> Meridian Freight is moving 400 TB of NFS file shares to Amazon EFS over a 10 Gbps Direct Connect link. Users keep working on the shares until a weekend cutover.
>
> Every file must be verified, and the cutover window is four hours.

What should the solutions architect use?

A. A Storage Gateway File Gateway in front of EFS

B. Custom rsync scripts on several servers

C. AWS DataSync: an initial full copy, scheduled incremental tasks, and a final incremental run at cutover with verification

D. AWS Transfer Family

◆ PAUSE AND DECIDE

What should the solutions architect use?

A File Gateway

Hybrid access to S3, not a migration to EFS.

B rsync scripts

Works, but custom, slower, and verification is yours to build.

C DataSync incremental

Bulk copy early, small final delta, built-in verification.

D Transfer Family

For SFTP and FTP clients, not bulk NFS migration.

THE SIGNAL

Large shares, live until cutover, verified: DataSync, full then incremental.

> Forty customs brokers send files to Meridian Freight's on-premises SFTP server, which is being retired. Brokers cannot change their software, credentials or the hostname they connect to.
>
> Files must land in Amazon S3.

What should the solutions architect implement?

A. An AWS Transfer Family SFTP server backed by S3, with brokers' keys imported and the hostname's DNS record pointed at it

B. Ask brokers to upload with the AWS CLI

C. AWS DataSync from the old SFTP server, kept running indefinitely

D. An EC2 instance running an SFTP daemon that copies to S3

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Transfer Family

Managed SFTP into S3; same hostname and keys for brokers.

B AWS CLI

Brokers cannot change their software.

C DataSync forever

Keeps the server that is being retired.

D Self-managed SFTP

Works, but an EC2 server to patch and scale.

THE SIGNAL

Partners' SFTP must keep working, data into S3: Transfer Family.

> Meridian Freight runs several DMS migrations with full load and change data capture. Load varies widely, from heavy initial loads to near-idle replication overnight.
>
> The team does not want to choose or resize replication instances.

What should the solutions architect recommend?

A The largest replication instance class for every task

B AWS Glue jobs instead of DMS

C Lambda functions polling the source tables

D AWS DMS Serverless replications

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Largest instances

Pays for peak capacity overnight.

B Glue

ETL, not log-based change data capture.

C Polling Lambda

Misses deletes and adds load to the source.

D DMS Serverless

Provisions and scales capacity automatically.

THE SIGNAL

DMS without sizing instances, load varies: DMS Serverless.

What to carry out of this module

- **Full copy, then changes**: downtime is only the final catch-up.
- **Engine change**: Schema Conversion first, then DMS full load plus CDC.
- **DataSync** for files, with incremental passes and verification.
- **Snowball is closed** to new customers; use Data Transfer Terminal.
- **Do the arithmetic**: about 10 TB a day per fully used Gbps.

> ON THE EXAM
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> Convert if needed, copy in bulk, stream the changes, cut over briefly.